Homology of the Sphere

Theorem. For $n \geq 1$,

$$H_q(S^n) \cong \begin{cases} \mathbb{Z} & q = 0 \text{ or } q = n, \\ 0 & \text{otherwise.} \end{cases}$$

Moreover,

$$H_q(D^n, S^{n-1}) \cong \begin{cases} \mathbb{Z} & q = n, \\ 0 & \text{otherwise.} \end{cases}$$

Step 1: Homology of S^0

The 0-sphere S^0 consists of two points. Hence it has two connected components. Therefore

$$H_0(S^0) \cong \mathbb{Z} \oplus \mathbb{Z}, \qquad H_q(S^0) = 0 \quad (q > 0).$$

Step 2: The Long Exact Sequence of a Pair

Consider the pair (D^n, S^{n-1}). The long exact sequence in homology gives

$$\cdots \to H_q(S^{n-1}) \xrightarrow{i_*} H_q(D^n) \to H_q(D^n, S^{n-1}) \to H_{q-1}(S^{n-1}) \to H_{q-1}(D^n) \to \cdots$$

Since the disk D^n is contractible, its homology groups are

$$H_q(D^n) = \begin{cases} \mathbb{Z} & q = 0, \\ 0 & q > 0. \end{cases}$$

Step 3: Computation of $H_0(D^n, S^{n-1})$

From the long exact sequence we obtain

$$H_0(S^{n-1}) \xrightarrow{i_*} H_0(D^n) \to H_0(D^n, S^{n-1}) \to 0.$$

For $n \geq 2$, both S^{n-1} and D^n are path connected. Hence

$$H_0(S^{n-1}) \cong H_0(D^n) \cong \mathbb{Z}.$$

The map i_* sends the generator of $H_0(S^{n-1})$ to the generator of $H_0(D^n)$, so it is surjective. By exactness we conclude

$$H_0(D^n, S^{n-1}) = 0.$$

Step 4: Computation of $H_1(D^n, S^{n-1})$

Consider the relevant part of the exact sequence:

$$H_1(D^n) \to H_1(D^n, S^{n-1}) \to H_0(S^{n-1}) \xrightarrow{i_*} H_0(D^n).$$

Since D^n is contractible,

$$H_1(D^n) = 0.$$

Thus

$$H_1(D^n, S^{n-1}) \cong \ker(i_*).$$

For $n \geq 2$, $i_* : \mathbb{Z} \to \mathbb{Z}$ is an isomorphism, hence

$$H_1(D^n, S^{n-1}) = 0.$$

For $n = 1$, the map $i_* : \mathbb{Z} \oplus \mathbb{Z} \to \mathbb{Z}$ sends (a, b) to $a + b$, whose kernel is $\mathbb{Z}$. Hence

$$H_1(D^1, S^0) \cong \mathbb{Z}.$$

Step 5: Hemispheres of S^n

Let

$$D^n_+ = \{x \in S^n : x_0 \geq 0\}, \qquad D^n_- = \{x \in S^n : x_0 \leq 0\}.$$

These are the upper and lower hemispheres of S^n. Both are homeomorphic to D^n.

Since D^n_- is contractible, the long exact sequence of the pair (S^n, D^n_-) gives

$$H_q(S^n) \cong H_q(S^n, D^n_-).$$

Step 6: Excision

By the excision theorem,

$$H_q(S^n, D^n_-) \cong H_q(D^n, S^{n-1}).$$

Hence

$$H_q(S^n) \cong H_q(D^n, S^{n-1}).$$

From the long exact sequence of the pair (D^n, S^{n-1}) we obtain

$$H_q(D^n, S^{n-1}) \cong H_{q-1}(S^{n-1}).$$

Thus we obtain the important relation

$$H_q(S^n) \cong H_{q-1}(S^{n-1}).$$

Step 7: Induction

Using the above relation repeatedly,

$$H_q(S^n) \cong H_{q-1}(S^{n-1}) \cong H_{q-2}(S^{n-2}) \cong \cdots$$

Eventually we reach $H_0(S^0)$.
Since $H_0(S^0) = \mathbb{Z} \oplus \mathbb{Z}$, tracking the degrees gives

$$H_n(S^n) \cong \mathbb{Z}$$

and all intermediate homology groups vanish.

Conclusion

Therefore,

$$H_q(S^n) \cong \begin{cases} \mathbb{Z} & q = 0 \text{ or } q = n, \\ 0 & \text{otherwise.} \end{cases}$$